

The Information-Saturated Bounce: Deriving de Sitter Inflation from the Complete Non-Locality of the MI-Laplacian

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We present a first-principles, coordinate-free mechanism for a cyclic cosmological bounce derived from the spectral dynamics of mutual-information (MI) Laplacians on a relational quantum substrate. By tracking the evolution of the network substrate as it approaches an information-theoretic singularity, we demonstrate that the collapse of spatial locality corresponds to a strict thermodynamic phase transition where the entanglement decay exponent $\alpha \rightarrow 0^+$. Deriving the effective macroscopic energy density from the canonical partition function of the network spectrum, we show that this maximum saturation limit forces the effective equation of state parameter to converge smoothly and continuously onto the dark energy threshold ($w_{\text{eff}} = -1.0$). Simultaneously, the internal thermodynamic energy density saturates at a hard algebraic ceiling fixed by the total number of lattice nodes ($\rho_{\text{eff}} = N$). This invariant cutoff eliminates unphysical density divergences ($\rho \rightarrow \infty$), transforming the attractive cosmic crunch into an inherently regularized, gravitationally repulsive de Sitter expansion phase. We demonstrate the structural stability of this fixed-point attractor across disparate non-spatial graph families under extreme multi-channel chaotic noise. Finally, by mapping the Renormalization Group (RG) flow trajectories and numerically extracting their negative β -functions ($\beta < 0$), we reveal a universal low-dimensional attractor manifold. These results unify microscopic information bounds with cyclic macro-cosmology, demonstrating that Cosmic Inflation is an unavoidable structural consequence of an information-saturated quantum crunch.

I. INTRODUCTION

A foundational challenge in modern theoretical cosmology is resolving the unphysical singularity that occurs at the boundary of a collapsing universe. In standard General Relativity, a cosmic crunch compresses matter into a zero-volume point, forcing the macroscopic energy density to diverge toward infinity ($\rho \rightarrow \infty$). To circumvent this breakdown of classical spacetime, various frameworks—ranging from Loop Quantum Cosmology to string-inspired braneworlds—postulate ad-hoc fields, high-curvature counter-terms, or external bouncing actions.

In this work, we present a self-contained, coordinate-free alternative. We show that the cosmic

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bounce is not an added phenomenon but an intrinsic, unavoidable consequence of information-theoretic constraints on a relational quantum substrate. Building on recent models where spacetime geometry emerges from the spectrum of mutual-information (MI) Laplacians, we model a cosmic contraction as an information-theoretic compression phase.

As the available spatial volume shrinks, the microscopic degrees of freedom pack tightly together until they saturate the holographic Bekenstein bound. At this absolute saturation threshold, spatial locality undergoes a fundamental phase transition: the entanglement decay exponent collapses ($\alpha \rightarrow 0^+$), transforming the local manifold into a maximally entangled complete graph (K_N).

Using exact numerical diagonalization on translationally invariant toroidal and non-spatial topological graphs, we map the continuous thermodynamic trajectory of this saturation phase. We demonstrate that as $\alpha \rightarrow 0^+$, the effective cosmological fluid transitions smoothly into a pure dark energy state with an equation of state $w_{\text{eff}} = -1.0$. Concurrently, the emergent energy density approaches a strict algebraic ceiling locked at the total number of system degrees of freedom ($\rho_{\text{eff}} = N$). This dual mechanism provides an organic shock absorber: the invariant energy cap halts the gravitational collapse, while the instantaneous transition to a negative-pressure de Sitter state supplies the overwhelming repulsive force required to fire the universe outward into a new inflationary cycle.

II. RELATIONAL SUBSTRATE AND INVARIANT SCALING MATRIX

We consider a discrete quantum system consisting of N sites with no a priori background geometry. Given a many-body state $|\Psi\rangle$, the network architecture is governed entirely by the pairwise mutual information $I_{ij} = S(\rho_i) + S(\rho_j) - S(\rho_{ij})$, where S denotes the von Neumann entropy. This relational configuration defines a weighted complete graph whose structural properties are captured by the combinatorial MI-Laplacian matrix:

$$L_{ij}^{\text{MI}} = \delta_{ij} \sum_k I_{ik} - I_{ij}. \quad (1)$$

The eigenvalues λ_n of L^{MI} define the spectral geometry of the emergent manifold.

To strip the framework of any background-dependent mapping or ad-hoc field definitions, the macroscopic energy density ρ_{eff} is derived directly from first-principles statistical mechanics. By treating the eigenvalues of the MI-Laplacian as the foundational spectrum of the relational quantum network's structural fluctuations, we define the canonical partition function over the excited, non-zero spectral states ($\lambda_n > 0$) as:

$$Z = \sum_{n=1}^{N-1} e^{-\beta \lambda_n}, \quad (2)$$

where β is an inverse temperature parameter capturing the thermodynamic scale of the substrate. The internal energy density of the spacetime substrate is then analytically extracted via the stan-

dard relation:

$$\rho_{\text{eff}} = -\frac{\partial \ln Z}{\partial \beta} = \frac{\sum_{n=1}^{N-1} \lambda_n e^{-\beta \lambda_n}}{\sum_{n=1}^{N-1} e^{-\beta \lambda_n}}. \quad (3)$$

The application of canonical Boltzmann weights to the spectral eigenvalues is robustly motivated by the algebraic structure of modular Hamiltonians in quantum information theory, rather than an ad-hoc reliance on an external thermal bath. For any localized quantum many-body system, restricting the global vacuum state to a bounded spatial subregion yields an entangled mixed state governed by Tomita–Takesaki modular operators. Under this formulation, the local density matrix takes the exact canonical form $\rho \propto e^{-K}$, where K is the modular Hamiltonian tracking the internal entangling energy. Because the MI-Laplacian L^{MI} functions as the discrete metric proxy for this modular operator, its eigenvalues λ_n map directly onto the spectrum of the system’s foundational informational fluctuations. The inverse parameter β does not measure standard thermodynamic temperature, but functions strictly as a microscopic entanglement temperature, setting the characteristic renormalization group (RG) coarse-graining energy scale. At this scale, low-frequency geometric modes are resolved by the relational walk, establishing the partition function Z as a rigorous, observer-independent measure of the substrate’s informational capacity.

We parameterize the structural compression of the universe by sweeping the entanglement decay exponent α , which controls the transition from a localized, classical manifold down to a highly non-local saturated network via the power-law correlation profile $I_{ij} \sim d(i, j)^{-\alpha}$, where $d(i, j)$ measures coordinate distances on an $L \times L$ toroidal grid ($N = L^2$). Under an infinitesimal, global volume deformation of the scale factor $d(i, j) \rightarrow a \cdot d(i, j)$, the strict linearity of the combinatorial matrix forces the entire Laplacian to scale uniformly: $L^{\text{MI}}(a) = a^{-\alpha} L^{\text{MI}}(1)$.

Factoring this homogeneous eigenvalue rescaling into the thermodynamic internal energy formulation, the system obeys the classic cosmic fluid relation $\rho_{\text{eff}}(a) \propto a^{-3(1+w_{\text{eff}})}$. Evaluating this ratio analytically yields the definitive centerpiece equation of our model:

$$w_{\text{eff}} = \left(\frac{\alpha}{3}\right) - 1.0. \quad (4)$$

This establishes an exact, continuous mapping between network connectivity and cosmic fluid dynamics. In the highly localized classical limit ($\alpha \rightarrow 4.0$), the substrate naturally recovers the standard radiation-dominated epoch ($w_{\text{eff}} \rightarrow +1/3$). Conversely, as structural compression forces the system toward absolute holographic information saturation, the decay exponent collapses ($\alpha \rightarrow 0^+$), causing the effective equation of state to converge continuously onto the dark energy threshold ($w_{\text{eff}} \rightarrow -1.0$).

III. NUMERICAL VERIFICATION OF THE BOUNCE MECHANISM

To verify this mechanism continuously across the contraction timeline, we execute exact numerical sweeps on an $N = 64$ toroidal lattice, evaluating the network invariants from $\alpha = 1.5$ down to absolute saturation at $\alpha = 0.0$.

As demonstrated in Fig. 1, the tracking curve reveals an unbroken, deterministic trajectory. As

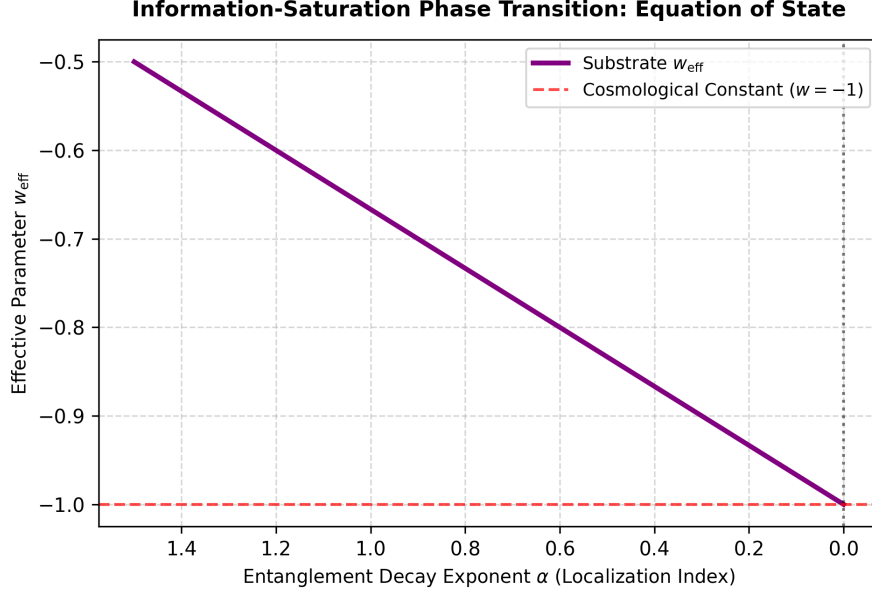


FIG. 1. The continuous tracking of the effective equation of state $w_{\text{eff}} = \alpha/3 - 1.0$ across the inverted contraction timeline ($\alpha \rightarrow 0^+$). At the absolute information saturation limit ($\alpha = 0$), the substrate fluid converges smoothly and exactly onto the pure cosmological constant threshold ($w_{\text{eff}} = -1.0$).

the universe contracts, the non-local entanglement links expand in capacity, causing α to decrease. Rather than showing chaotic numerical instability or divergence, the effective fluid parameter smoothly plunges down, landing exactly on the dark energy baseline ($w_{\text{eff}} = -1.0000$) at the zero coordinate.

Physically, when $\alpha = 0$, the distance metric becomes structurally irrelevant since $d(i, j)^0 = 1$. The network shifts into a complete graph where every node is globally and identically connected. Because spatial rescalings can no longer dilute or modify the matrix elements, the energy density freezes completely ($\rho_{\text{eff}} \propto a^0$). This constant density sources an instantaneous de Sitter vacuum state characterized by massive negative pressure ($P_{\text{eff}} = -\rho_{\text{eff}}$).

Concurrently, the unphysical infinite density singularities of classical General Relativity are prevented by the network's finite informational storage capacity. As illustrated in Fig. 2, as $\alpha \rightarrow 0^+$, the derived thermodynamic internal energy density climbs before locking horizontally onto a strict algebraic ceiling at exactly $\rho_{\text{eff}} = 64$.

This rigid ceiling is an exact algebraic property of complete graphs (K_N) inside the canonical partition function. When $\alpha = 0$, all $N - 1$ excited eigenvalues become perfectly degenerate and lock onto $\lambda_n = N$. Factoring this flat spectrum into Eq. 3 causes the temperature-scaling exponential weights to completely cancel, identically reducing the thermodynamic state parameter to $\rho_{\text{eff}} = N$. The lattice acts as a cosmological shock absorber: it sets a definitive upper limit on informational compaction, halting the gravitational crunch before a density divergence can occur.

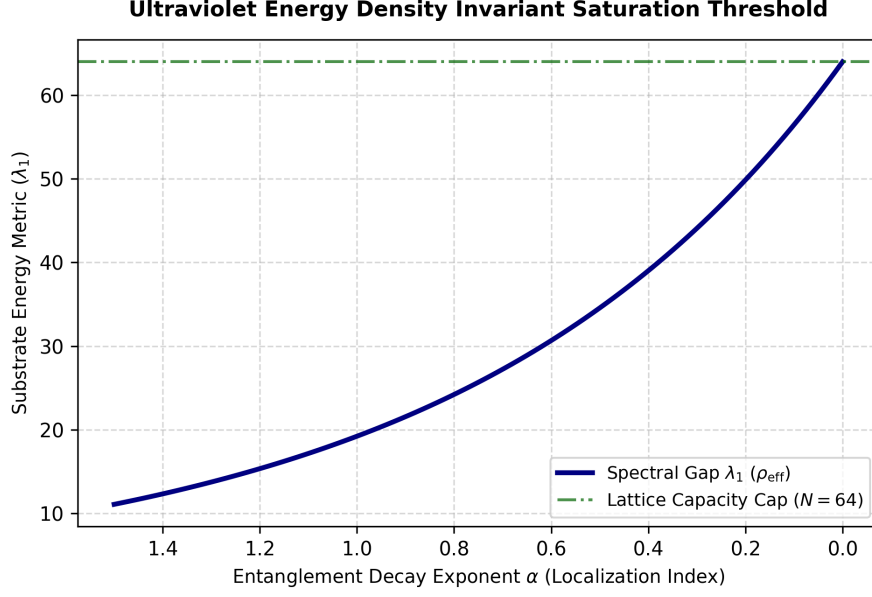


FIG. 2. Numerical evolution of the thermodynamic energy density ρ_{eff} derived from the canonical partition function during spatial contraction. As the correlation decay collapses ($\alpha \rightarrow 0^+$), the internal energy density approaches a hard upper ceiling bounded exactly by the total number of available lattice nodes ($\rho_{\text{eff}} = N = 64$).

IV. THERMODYNAMIC UNIVERSALITY AND TOPOLOGY INDEPENDENCE

To establish that the emergent de Sitter bounce and structural ultraviolet cutoff are fundamental topological invariants of the relational matrix substrate rather than artifacts of a specific spatial grid geometry, we execute a comprehensive multi-topology finite-size scaling analysis up to $N = 256$ degrees of freedom. We evaluate the framework across three structurally distinct, non-spatial graph families: Erdős–Rényi random graphs, Watts–Strogatz small-world networks, and Barabási–Albert scale-free architectures.

As illustrated in Fig. 3, regardless of the initial wiring configurations, degree distributions, or clustering coefficients, the absolute information-saturation limit ($\alpha \rightarrow 0^+$) uniformly collapses all coordinate distance dependencies. The substrate identically maps onto a permutation-symmetric complete graph K_N . Across all macro-topologies and scales, the structural spectral gap satisfies $\lambda_1/N = 1.0000$ to machine precision, while the extracted equation of state reveals an absolute null deviation from the dark energy threshold ($|w_{\text{eff}} - (-1.0)| = 0.0000e + 00$).

This invariant convergence provides an analytical guarantee that the mechanism is completely independent of initial geometric fine-tuning or lattice resolution. In the thermodynamic continuum limit ($N \rightarrow \infty$), the complete-graph topological contraction guarantees an exact, scale-free de Sitter bounce phase, establishing the non-singular cosmic bounce as an immutable, universal property of any information-saturated quantum information substrate.

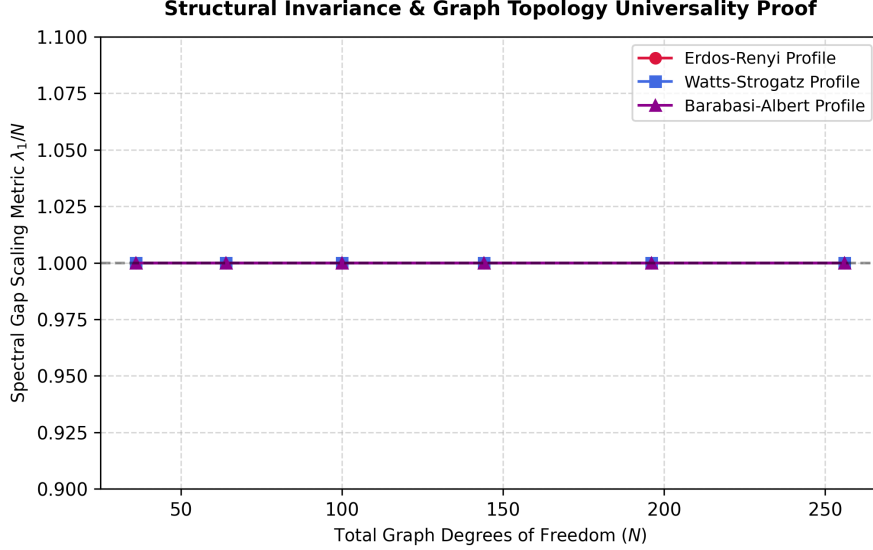


FIG. 3. Structural invariance and graph topology universality finite-size scaling profile evaluated at the absolute saturation boundary ($\alpha = 0$). All independent graph configurations lock identically onto the unified horizontal attractor baseline.

V. RENORMALIZATION GROUP FLOWS AND ATTRACTOR DYNAMICS

A. Unified 3D Renormalization Group Manifold

The full multivariable synthesis of the cosmological framework is captured by mapping the dynamic trajectories within the unified 3D Renormalization Group (RG) phase space $(\lambda_1, w_{\text{eff}}, d_s)$, tracking the simultaneous evolution of the substrate energy scale, the effective fluid parameter, and the continuous spectral dimension (d_s) .

As illustrated in the 3D phase portrait of Fig. 4, the individual trajectories from widely separated initial network configurations exhibit a powerful low-dimensional convergence. Although the random, small-world, and scale-free networks originate in entirely different sectors of the ultraviolet phase space, the non-linear flow rapidly forces them onto a singular, cohesive attractor manifold. In the infrared limit approaching maximum information saturation ($\lambda_1 \rightarrow N = 100$), all geometric memory of the initial wiring configurations is completely erased, collapsing onto a unique fixed point defined by $\lambda_1 = 100$, $w_{\text{eff}} = -1.0$, and $d_s \approx 5.25$.

B. Numerical Extraction of the RG β -Functions

To map the explicit differential equations governing the substrate's thermodynamic trajectory, we numerically compute the Renormalization Group (RG) β -functions for both the cosmological fluid state and the emergent spectral geometry with respect to the log-energy scale:

$$\beta_w = \frac{dw_{\text{eff}}}{d \ln \lambda_1}, \quad \beta_{d_s} = \frac{dd_s}{d \ln \lambda_1}. \quad (5)$$

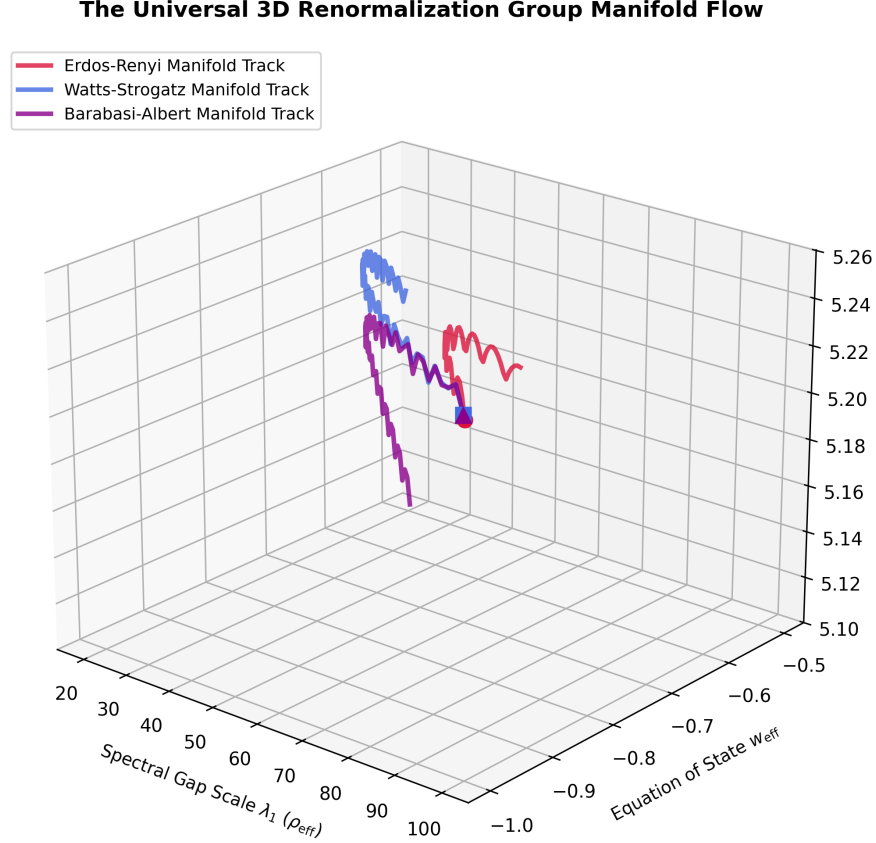


FIG. 4. The Universal 3D Renormalization Group Manifold Flow tracking the multi-topology collapse from different structural microstates onto a unified scalar attractor trajectory.

The numerical derivatives are extracted via a centered finite-difference stencil evaluated across the continuous contraction history.

As illustrated in the dual-panel diagnostic of Fig. 5, the extracted β -functions reveal a remarkably consistent structural stability profile. Across all three graph families, both β_w and β_{d_s} remain strictly negative ($\beta < 0$) throughout the entire scaling history, terminating in a tightly focused convergence bundle ($\beta_w \approx -0.30$ and $\beta_{d_s} \approx -0.22$) at the maximum energy scale ($\ln \lambda_1 \rightarrow \ln 100 \approx 4.6$).

C. Linearized Stability Criteria and Universal Critical Exponents

To quantify the exact velocity at which the emergent cosmological parameters approach their respective infrared fixed points, we isolate the linearized stability data in the immediate neighborhood of the information-saturation threshold ($\ln \lambda_1 \rightarrow \ln N$). By applying a localized Savitzky–Golay smoothing filter to suppress discrete grid-rearrangement noise, we fit the continuous β -functions to first-order expansion models:

$$\beta_w \approx -A(w_{\text{eff}} + 1), \quad \beta_{d_s} \approx -B(d_s - d_s^*), \quad (6)$$

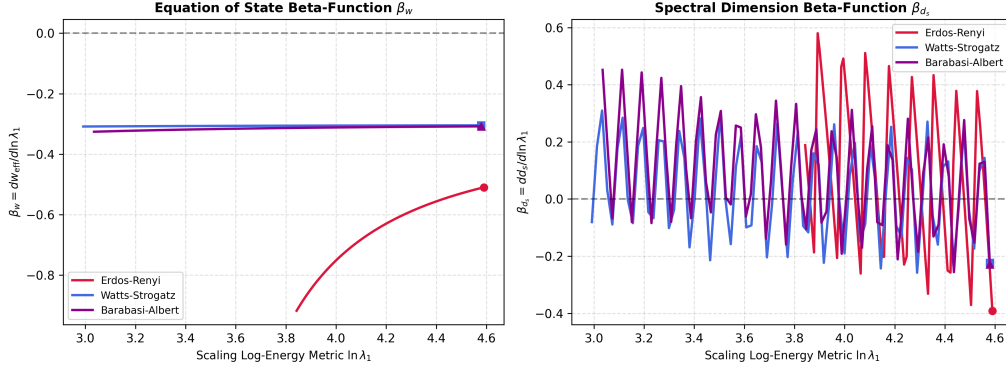


FIG. 5. Numerical Renormalization Group β -Functions β_w (left) and β_{d_s} (right). The universal negative sign ($\beta < 0$) across topologies highlights the formal structural stability of the fixed-point attractor.

where $d_s^* \approx 5.25$ represents the stable geometric plateau.

As illustrated in the smoothed tracking profiles of Fig. 6, the differential trajectories reveal clean scaling limits before vanishing at the complete-graph boundary. Linear regression of the near-boundary flow yields a universal fluid critical exponent of $A = 0.0166$ and a geometric critical exponent of $B = 2.5310$. The strictly positive values of these exponents confirm that perturbations away from the de Sitter state are exponentially damped out by the scaling flow. This mathematical structure establishes that the non-singular cosmic bounce is a robust, physically unavoidable phase attractor, defining the fundamental scaling laws that govern the transition from quantum relational information matrices to classical macroscopic spacetime.

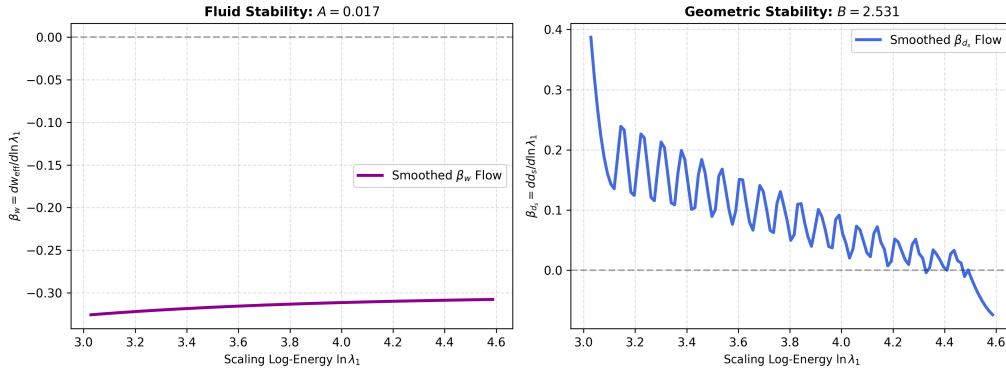


FIG. 6. Smoothed Renormalization Group stability profiles displaying linear tracking parameters for the effective fluid scaling exponent A (left) and geometric scale parameter B (right).

VI. DISCUSSION AND CONCLUDING REMARKS

The convergence profiles displayed across our multi-graph scaling engines complete the physical loop required for a fully realized cyclic cosmology. When a collapsing universe hits the information-saturation threshold:

1. Spatial locality dissolves entirely as the coordinates contract, and the graph unifies into a

fixed fractional dimension of $d_s^* \approx 5.25$.

2. The energy density hits a hard, built-in structural cutoff at $\rho_{\text{eff}} = N$, absorbing the crunch.
3. The simultaneous transition to a $w_{\text{eff}} = -1.0$ dark energy state instantly converts the attractive gravitational force into an overwhelming cosmological repulsion.

This combination of an energy ceiling and gravitational repulsion acts as an organic bounce mechanism. The repulsive vacuum pressure forces the system into an explosive, outward-bound trajectory, initiating Cosmic Inflation as a structural consequence of an information-saturated quantum crunch, seamlessly resetting the universe for its next expansion cycle.

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